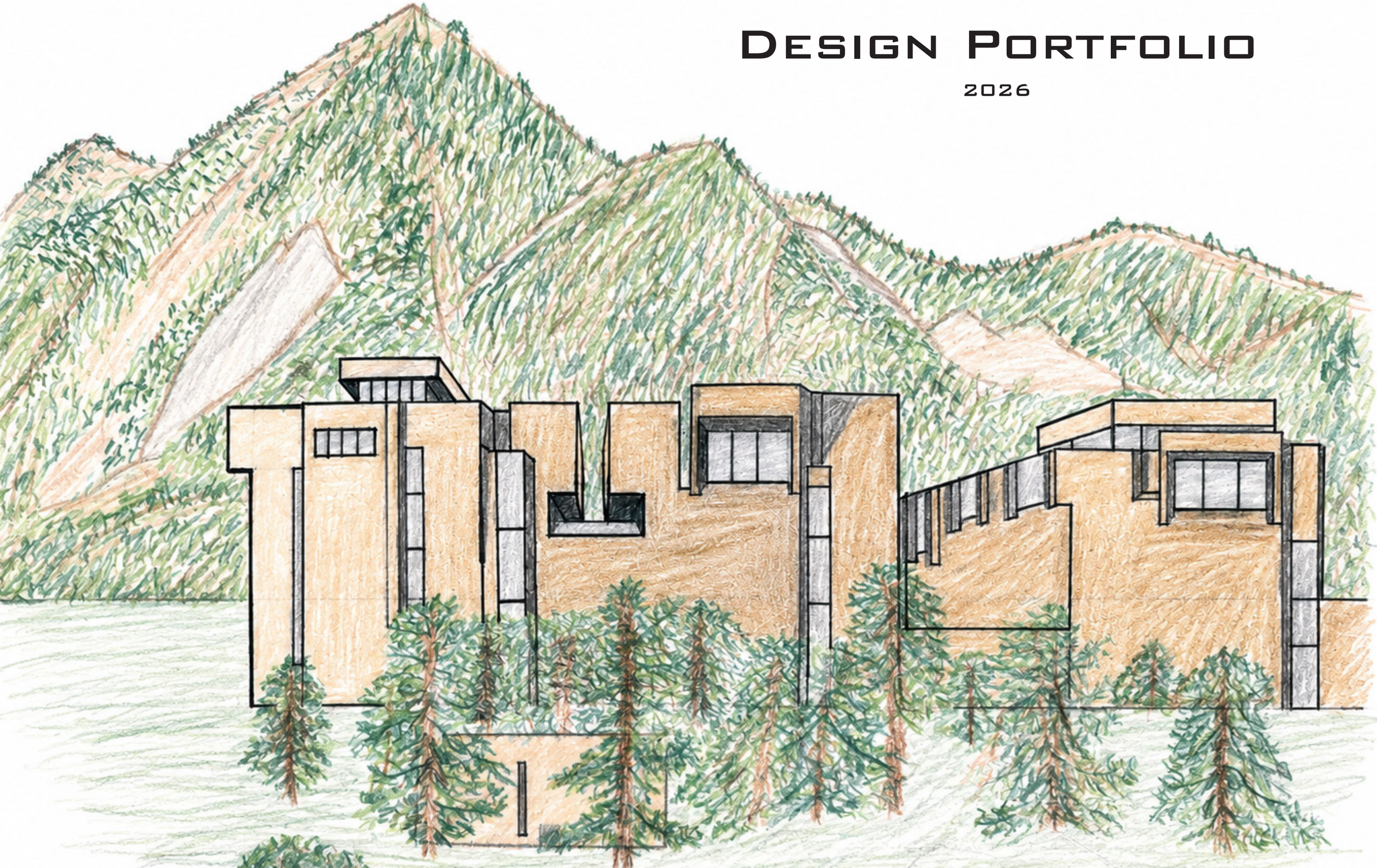
Design Portfolio

2026



Tyler Dove

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j.tyler.dove@gmail.com

Hey

I am a junior Civil Engineering student with a minor in Digital Fabrication, interested in pursuing architecture and design. My engineering background gives me a broad understanding of how things are built and shapes the way I approach design challenges. Through my studies and professional experiences, I have explored a range of projects and disciplines, developing my own approach to design and an appreciation for both the technical and creative sides of the process.

Learn More

Work Experience

Vanderbilt University - Graduating Spring 2028

- 3.9 GPA
- Member of Vanderbilt's chapters of the American Society of Civil Engineers and the student Architecture Society

Atkinson-Noland & Associates Engineering Design Internship - Summer 2025

- Contributed to historic preservation projects through Design modeling, documentation, and plan markups.
- Conducted site investigations and assessments of historic and commercial buildings.

Impressio Tech Product Design and Engineering Intern - Summer 2026

- Collaborated with design engineers throughout the product development process, from concept generation to production.
- Contributed to design iteration, prototyping, testing, and documentation across the product lifecycle

Negative Design CO. Product Design and Marketing Intern – Summer 2026

- Worked to develop interior Design Products
- Enhanced graphic design skills by creating and running Digital Ad Campaigns

Skills

- CAD Design:
  - Revit
  - AutoCAD
  - SketchUp
  - Solidworks
  - Civil 3D
  - Fusion 360
- InDesign
- Bluebeam
- SLS and SLA 3D Printing

Common Ground 03

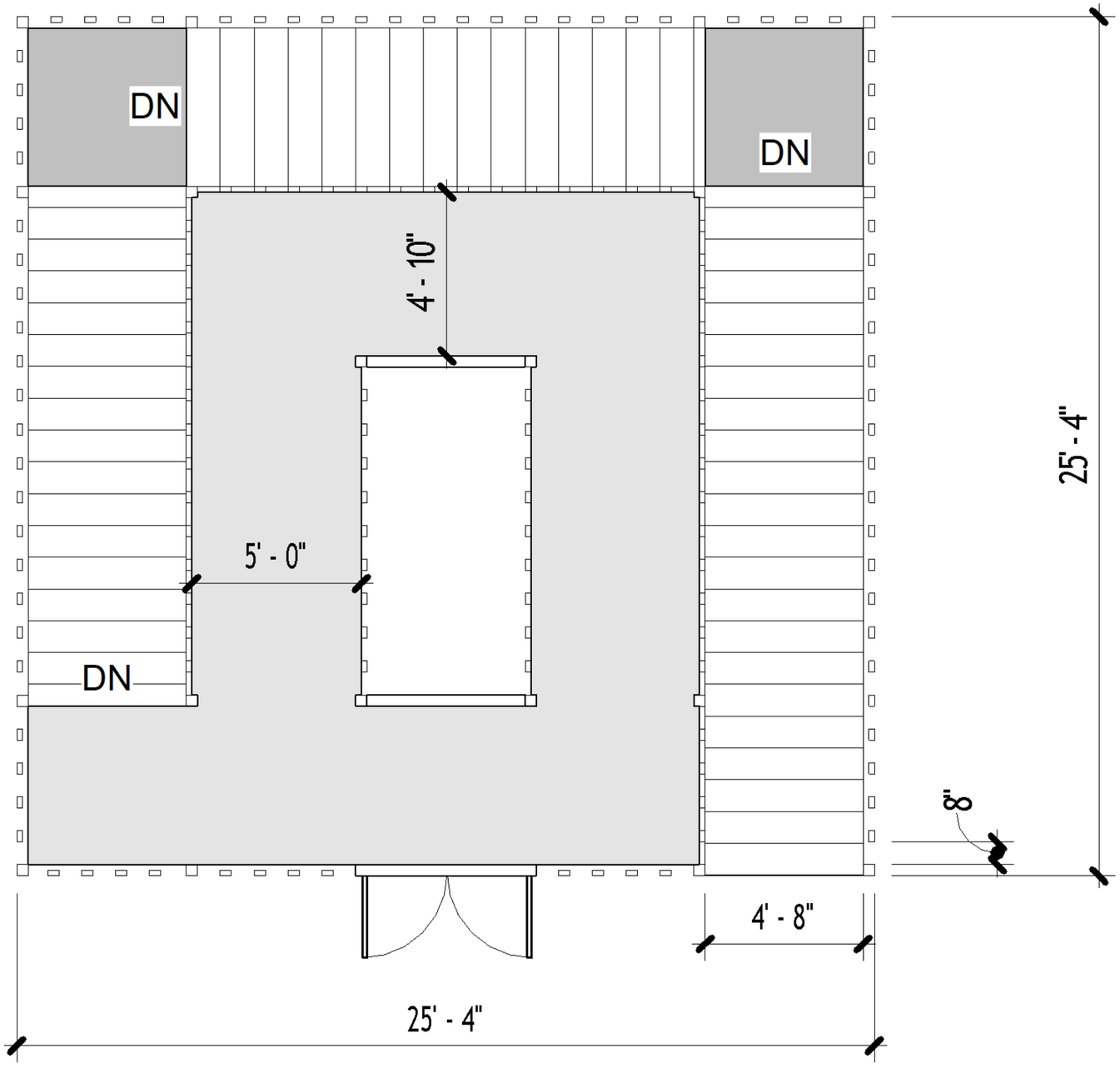
Harmony 07

Back of the Napkin 11

Island of Misfit Toys 15

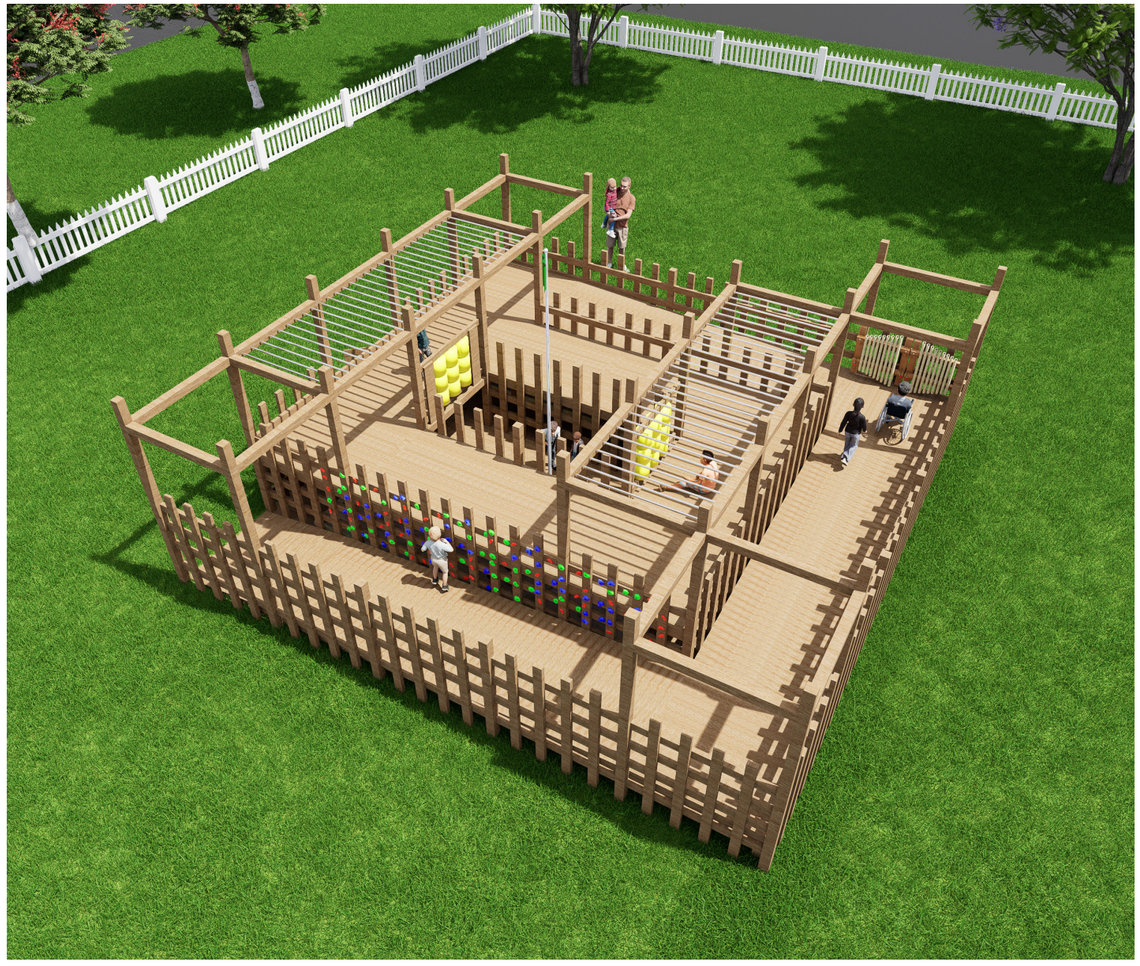
Common Ground

A Challenge in Accesibility for the students at the Susan Gray School



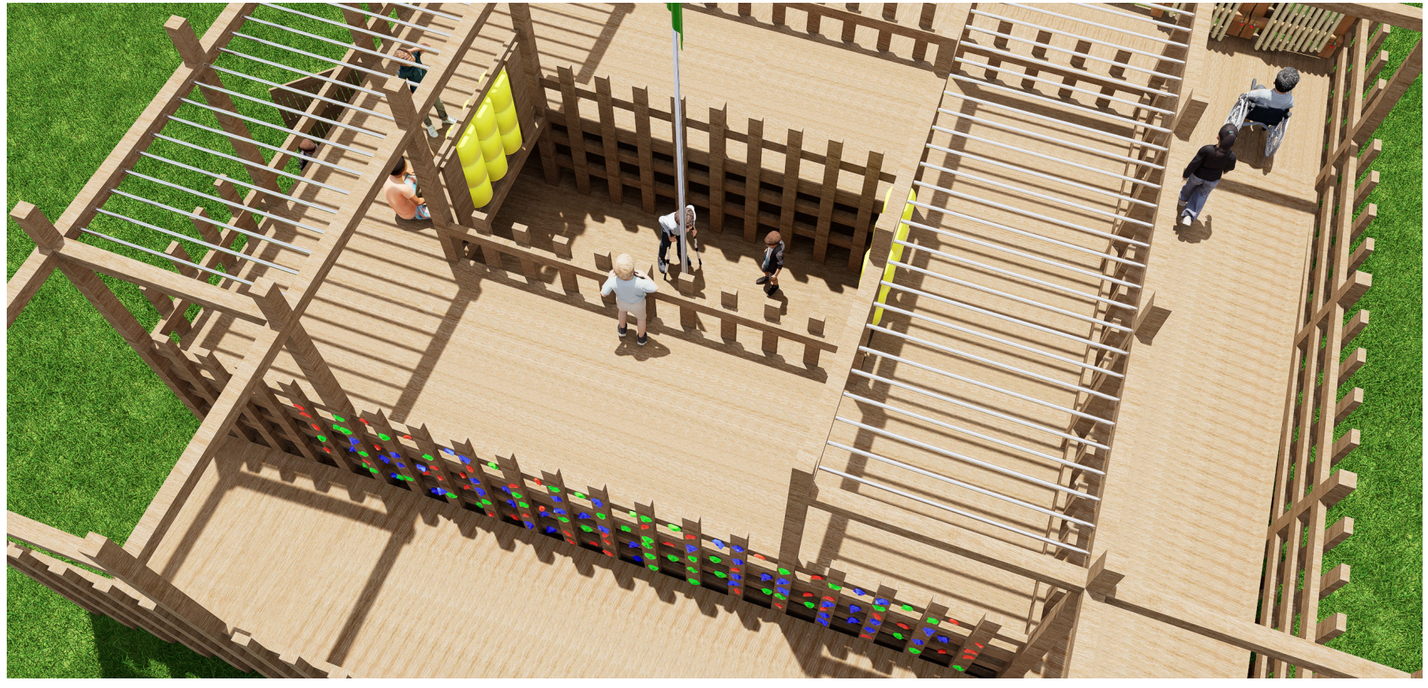
Designed For Simplicity of construction while maximizing accesibility with wide walkways and an accesibility ramp incorporated into the castle

the space encourages sensory stimulation and movement while the low height of the structure, 3ft, keeps play at a comfortable scale.



The playground transforms the idea of a castle into a space for exploration and imagination. Timber walls, elevated platforms, and covered walkways create a series of connected spaces that encourage students to move through, around, and within the structure.

Interior spaces are designed to feel playful and immersive, using changes in light, scale, and enclosure to create moments of discovery.



It wouldn't be a castle without a flag.

Built around a center courtyard that encourages collaboration through open space, elements like the flag pole encourage interactions between students of different physcual abilities.

The rock wall, soundmakers, castle flag, monkey bars, and tic-tac-toe boards each offer a different way to engage with the structure, creating opportunities for climbing, reaching, listening, and hands-on play.

The Harmony Project

Building a Sustainabile future

A facility that works alongside its community
rather than imposing upon it.

1-4: MAIN CAMPUS BUILDINGS
5: EXISTING BUILDING ON SITE, CONVERTED TO SECUIRTY CENTER
6: EXISTING BUILDING ON SITE, CONVERTED TO GREENHOUSE
7: SUBSTATION
8: DETENTION PONDS
A: ADDED PAVEMENT
B: CEMENT WALKWAYS
C: GRAVEL FIRE ROAD

BIG BROWN RIVER

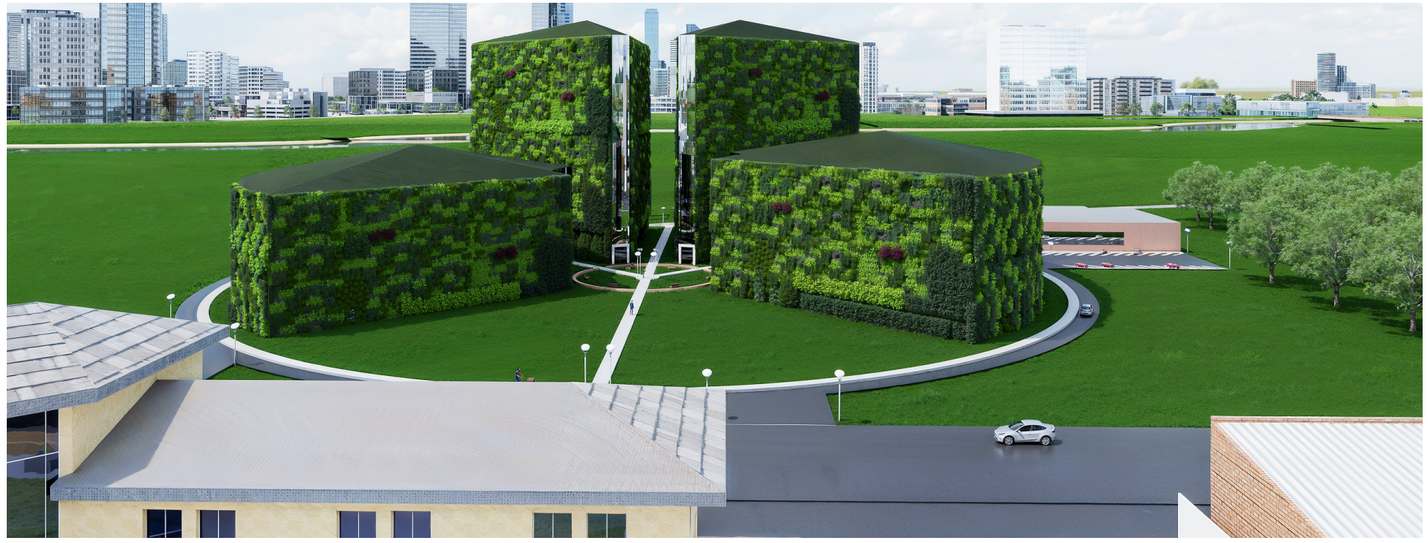
(306) (308) (310) (312) (314) (316) (318) (320) (322) (324) (326) (328) (330) (332)

1 2 3 4 5 6 7 8 A B C

ASPHALT CONCRETE

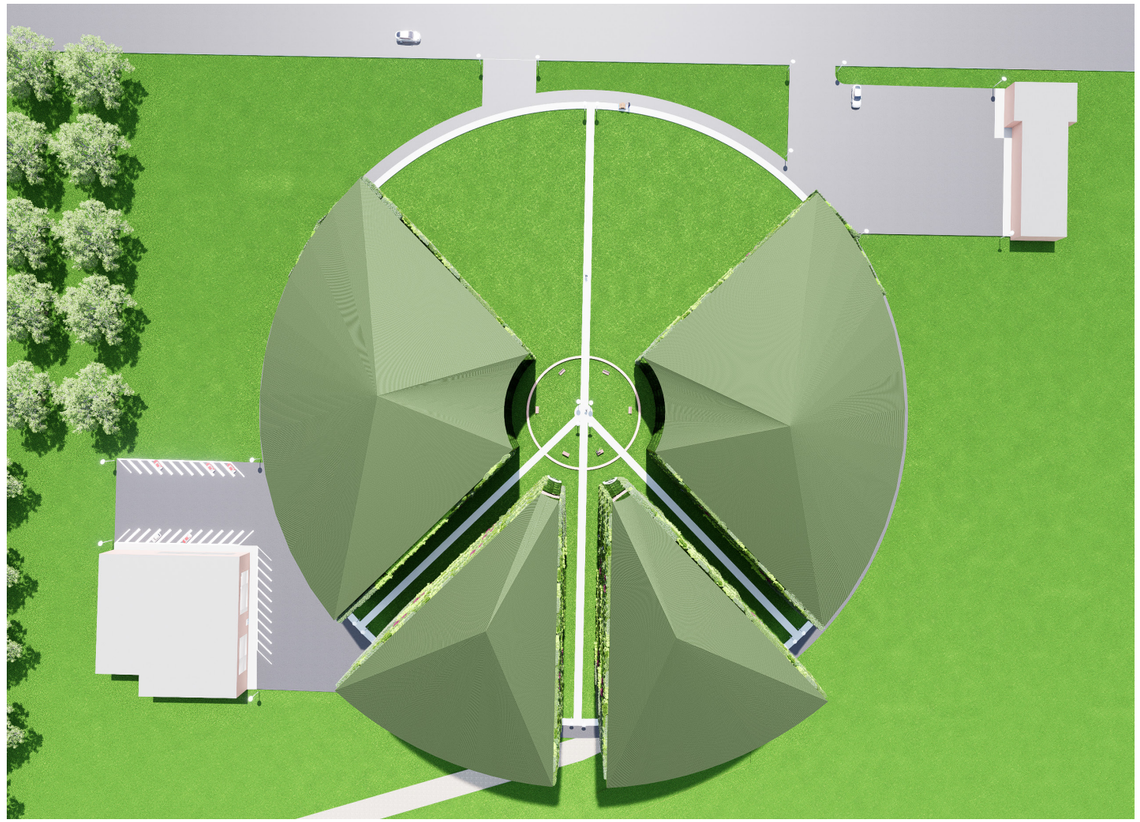
200' 100' 0' 100' 200'

08



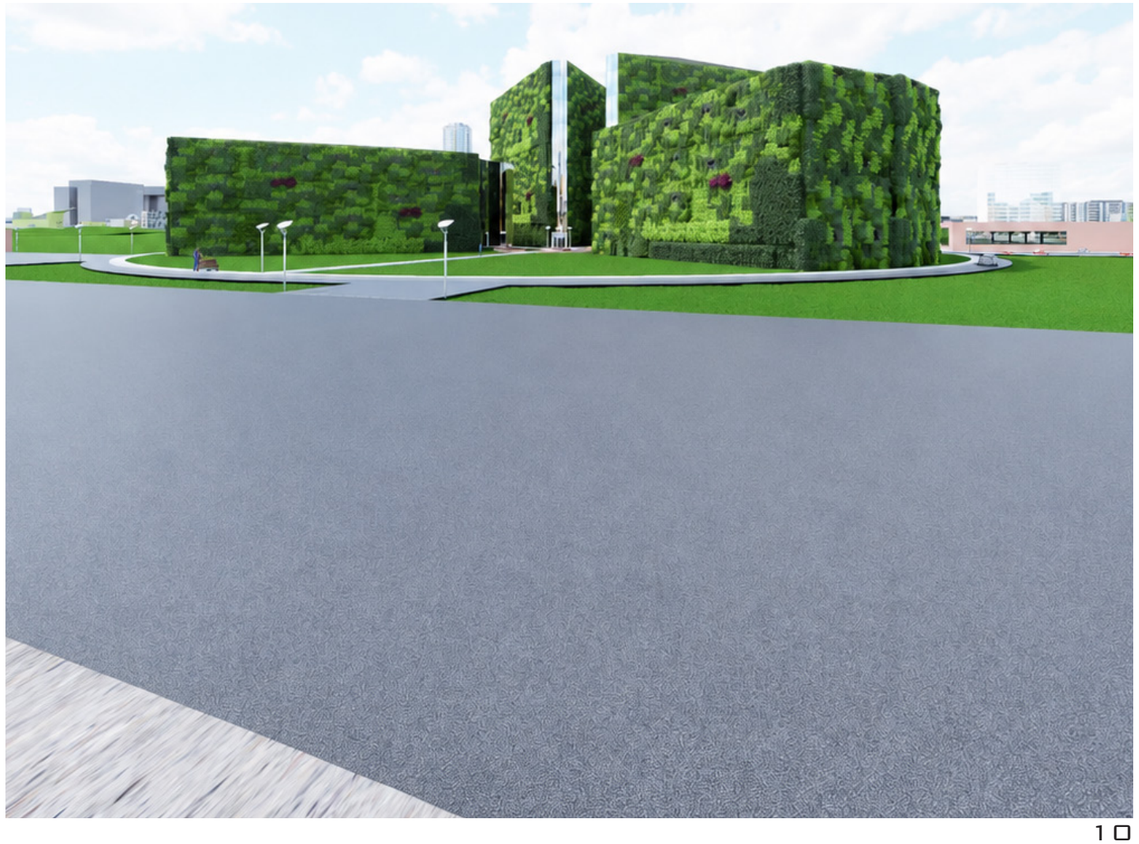
Data centers are shaping the future of technology, but must that future come at the expense of the communities around them? Harmony challenges this notion, reimagining data center infrastructure as an extension of its natural surroundings. Harmony brings technology and nature into balance.

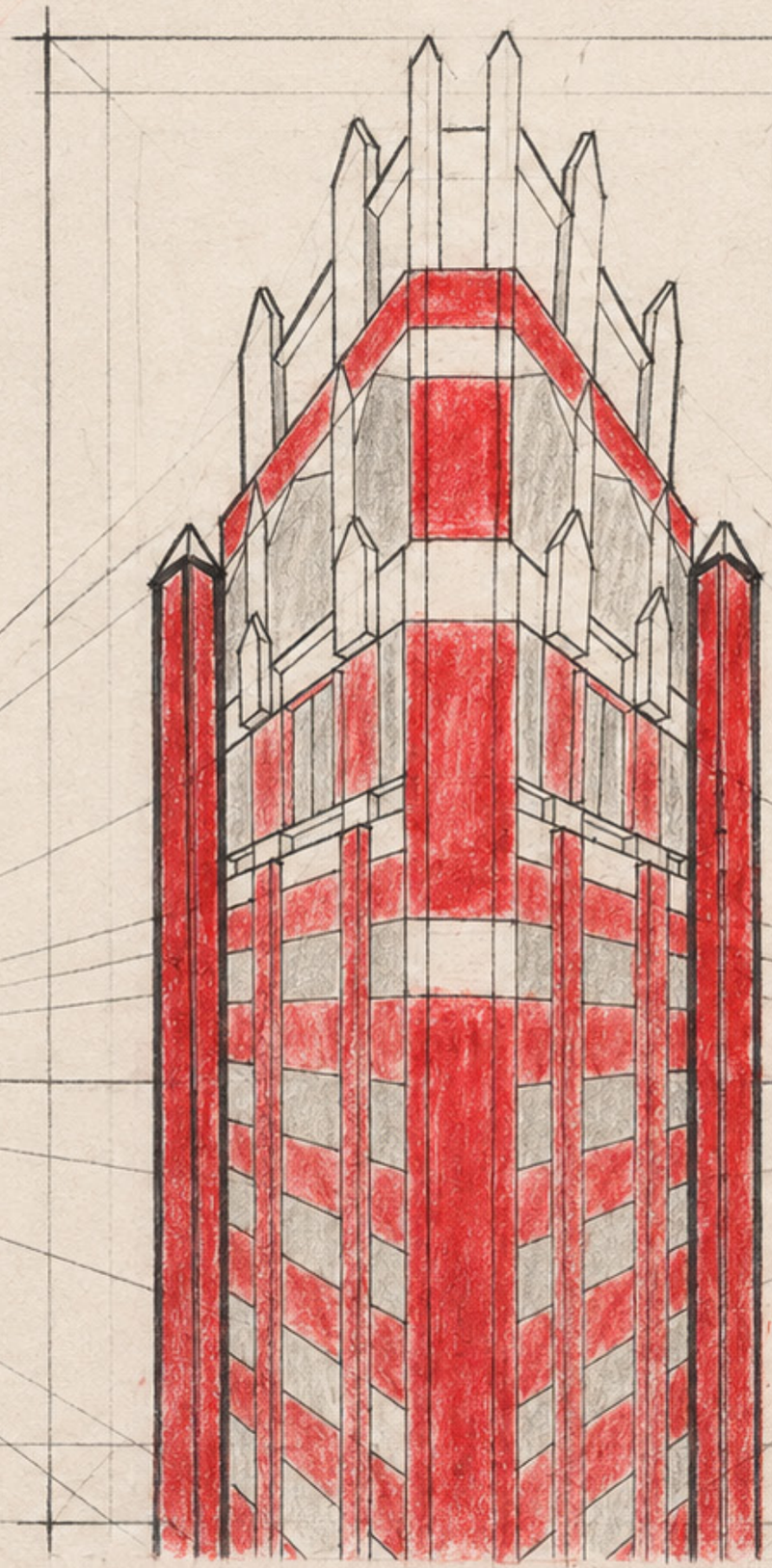
visually disruptive infrastructure, such as the electrical substation, are hidden by the site environment, transforming industrial elements into an extension of the landscape.



The peace-sign geometry serves as the central organizing concept of Harmony, symbolizing the project's commitment to coexistence, balance, and unity. More than a visual gesture, the design reinforces the idea that technological progress, environmental responsibility, and community well-being do not have to exist in opposition.

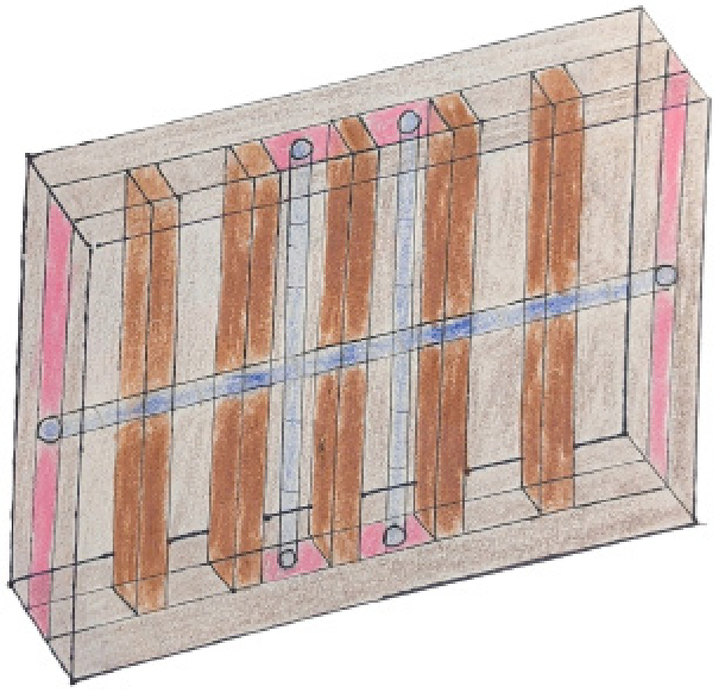
Community impact informed every aspect of the design. The goal was to minimize the visual and physical footprint of the facility while preserving the character of its surroundings.

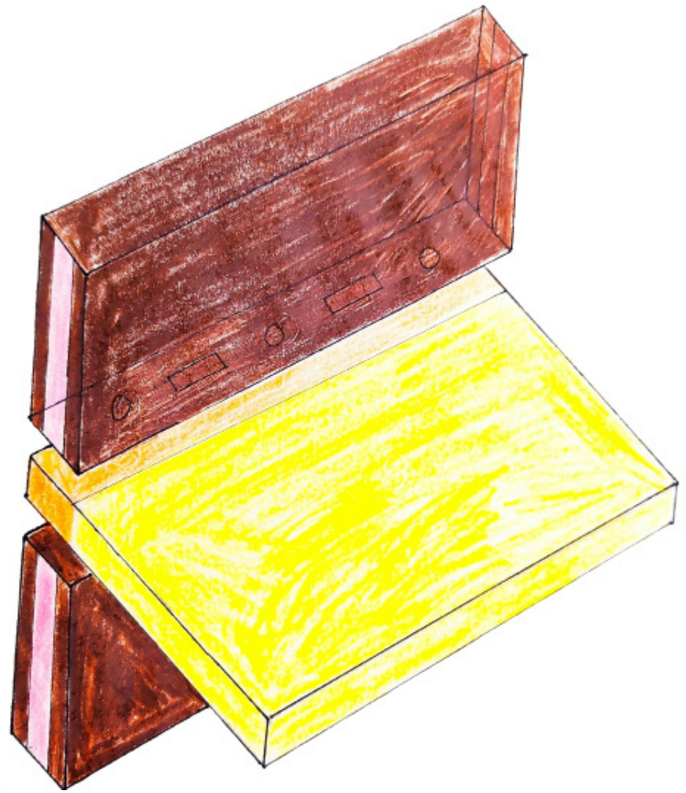


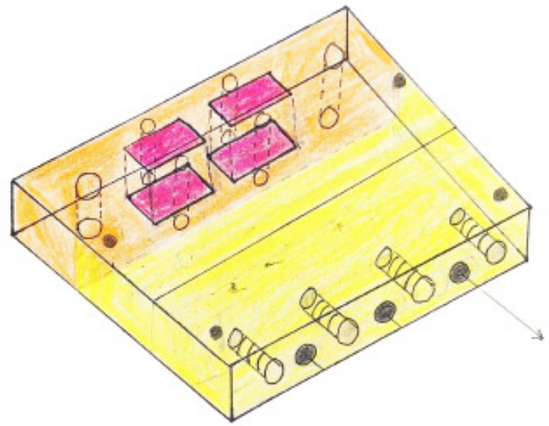


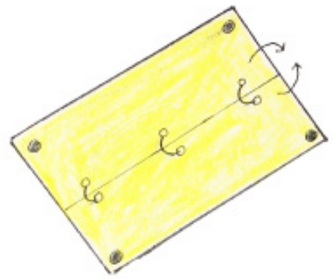
BACK OF THE NAPKIN

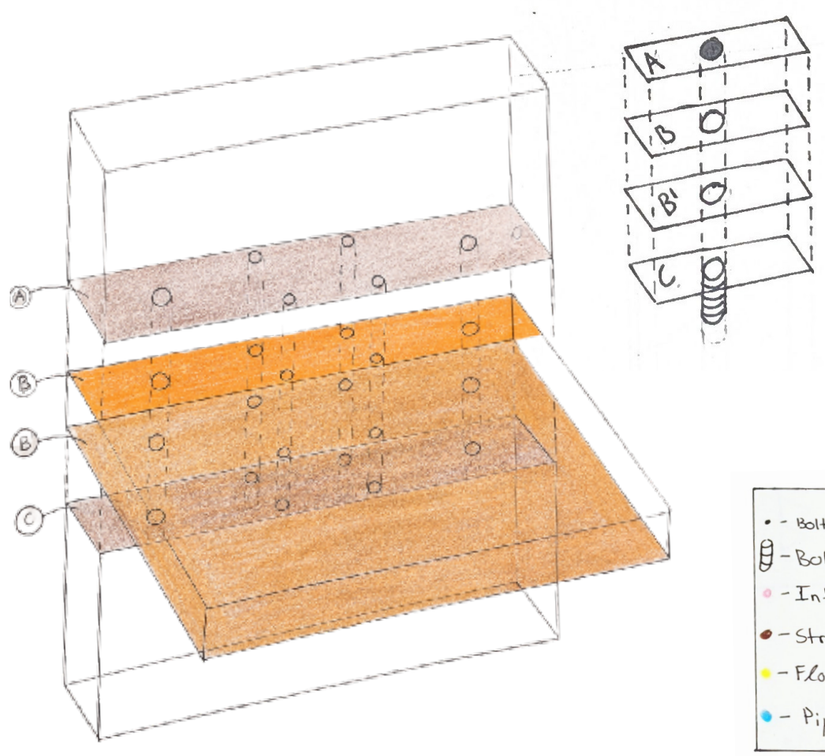
SKETCHING
CONCEPT
DRAWINGS

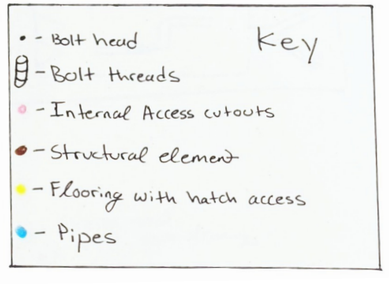


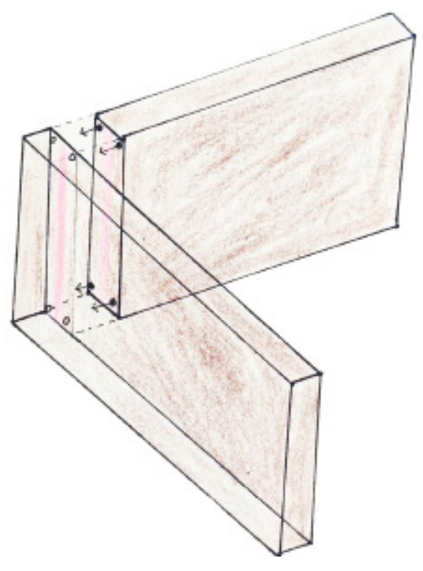












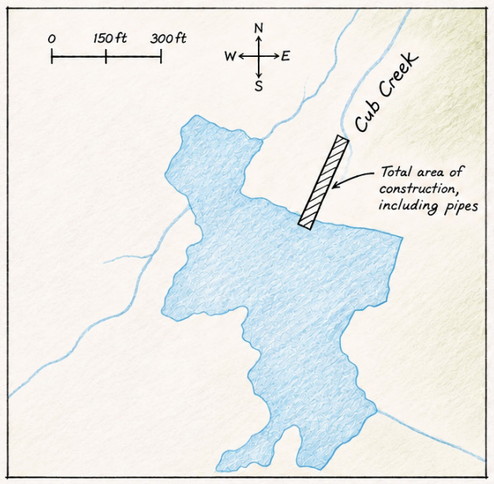
PanelLinx

These sketches explore the development of a standardized modular panel system inspired by the simplicity of LEGO bricks clicking together. The goal was to simplify traditional construction through a repeatable connection system that allows individual wall and floor panels to quickly lock together on-site. I explored how structural fasteners, integrated utilities, and accessible connection points could be incorporated into each panel while maintaining a simple assembly process.

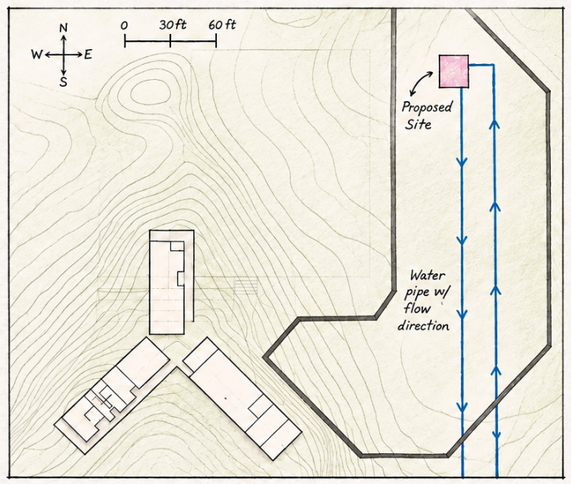
FloodLab - A Manmade Watershed
flash Flood Research Concept

The Tennessee WET Center sought new ways to educate engineers and the public about stormwater and flood resilience. Inspired by Tennessee's flooding challenges and the natural filtration of bioswales, I developed a concept for a closed-loop flash flood testing system with eight channels comparing different surfaces and green infrastructure strategies. Utilizing available site plans and creating concept sketches, I helped translate my idea into a buildable design. Leveraging digital design tools and AI prompt engineering to enhance my original sketches helped bring them to life and communicate the concept more effectively.

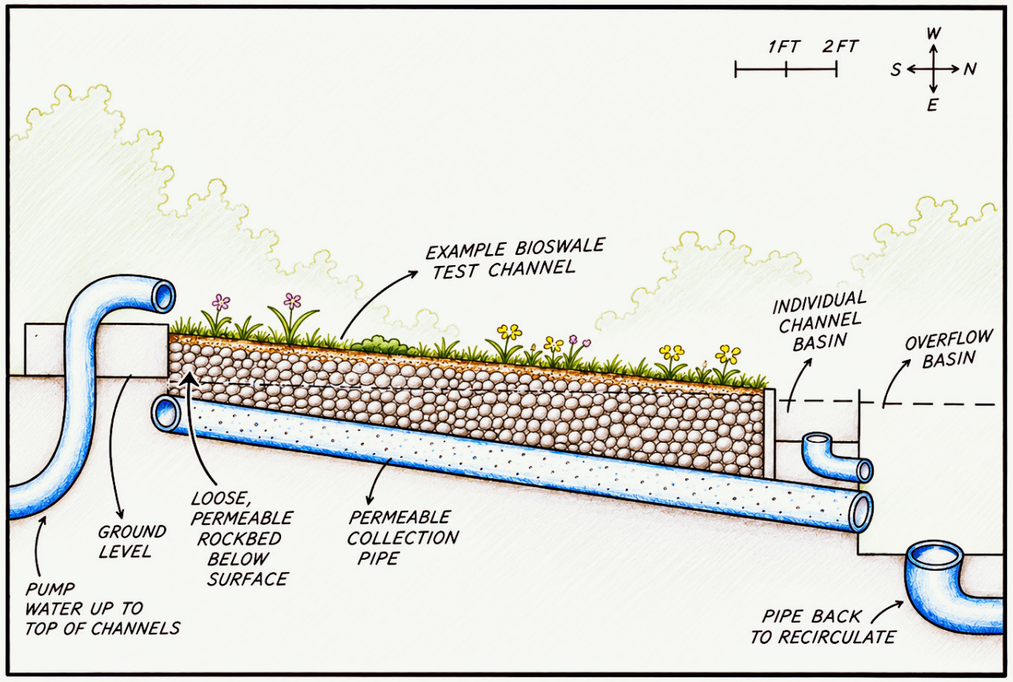
LONE OAKS FARM AREA OVERVIEW



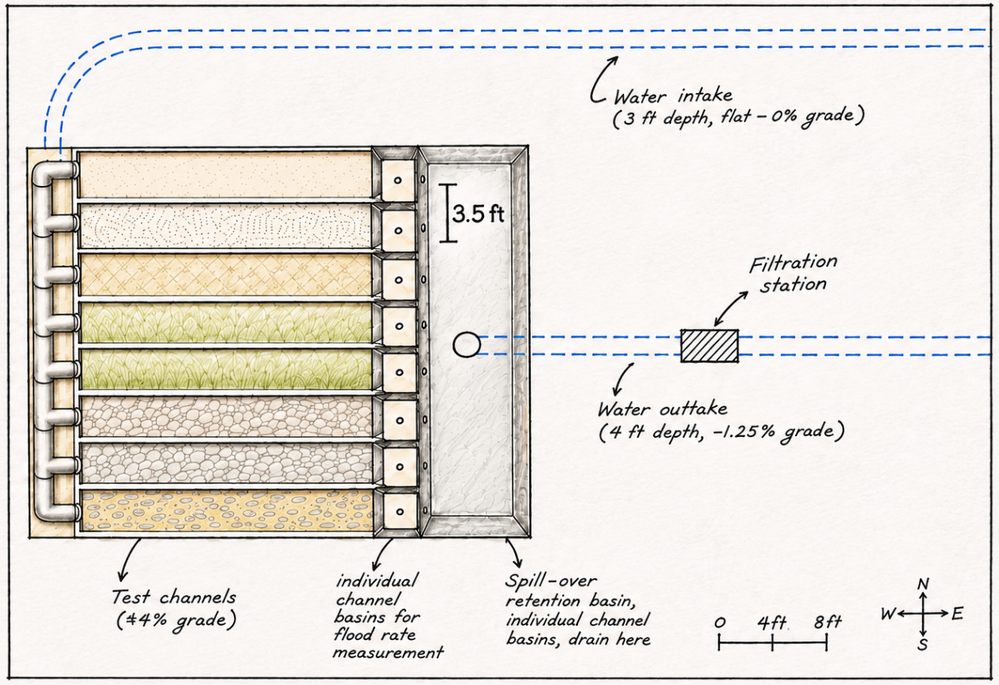
WET CENTER SITE PLAN OVERVIEW

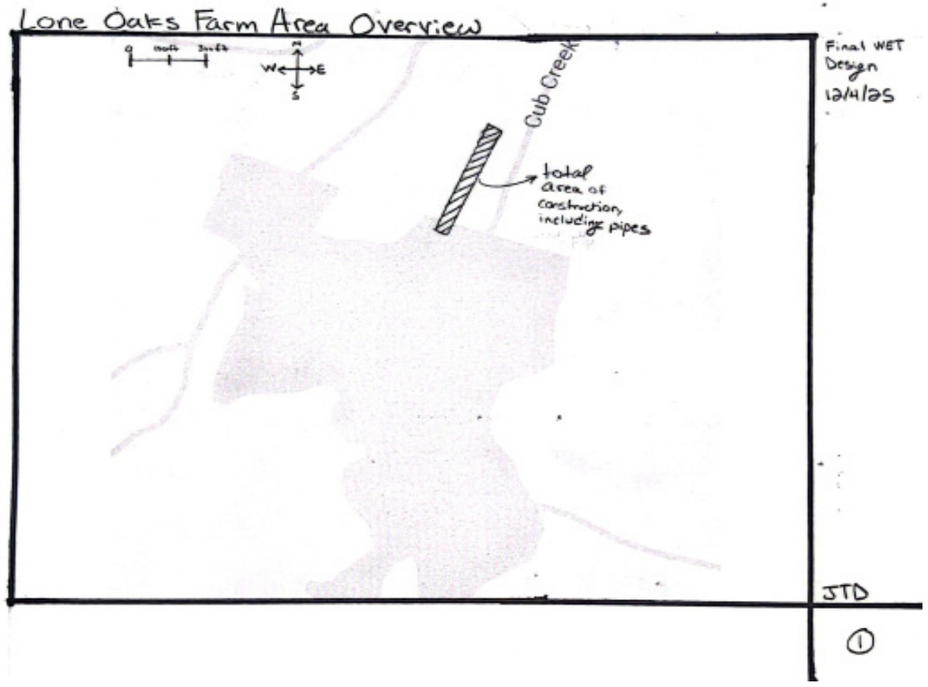


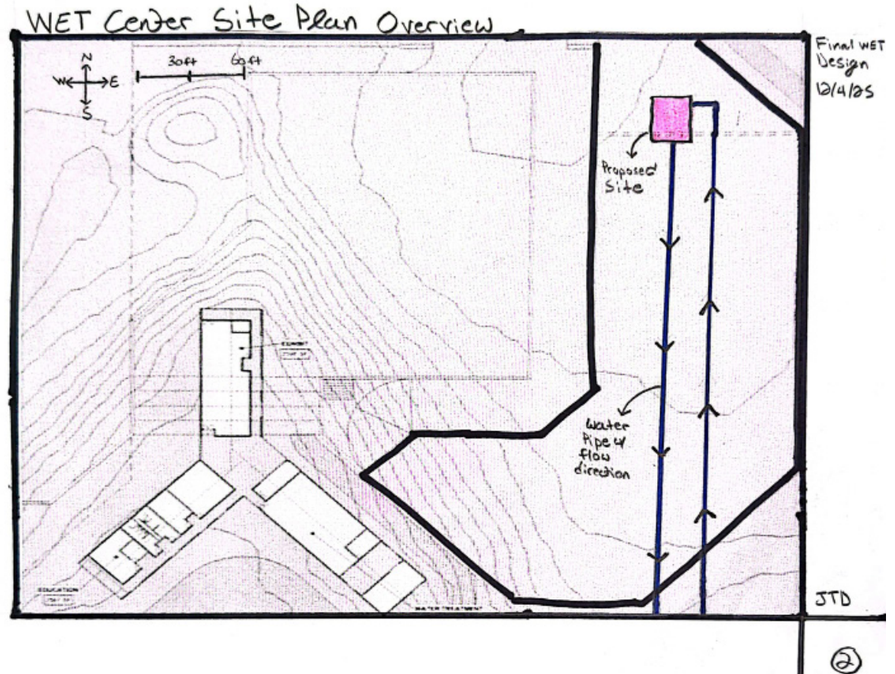
SIDE-ON TEST CHANNEL SYSTEM – WEST FACING

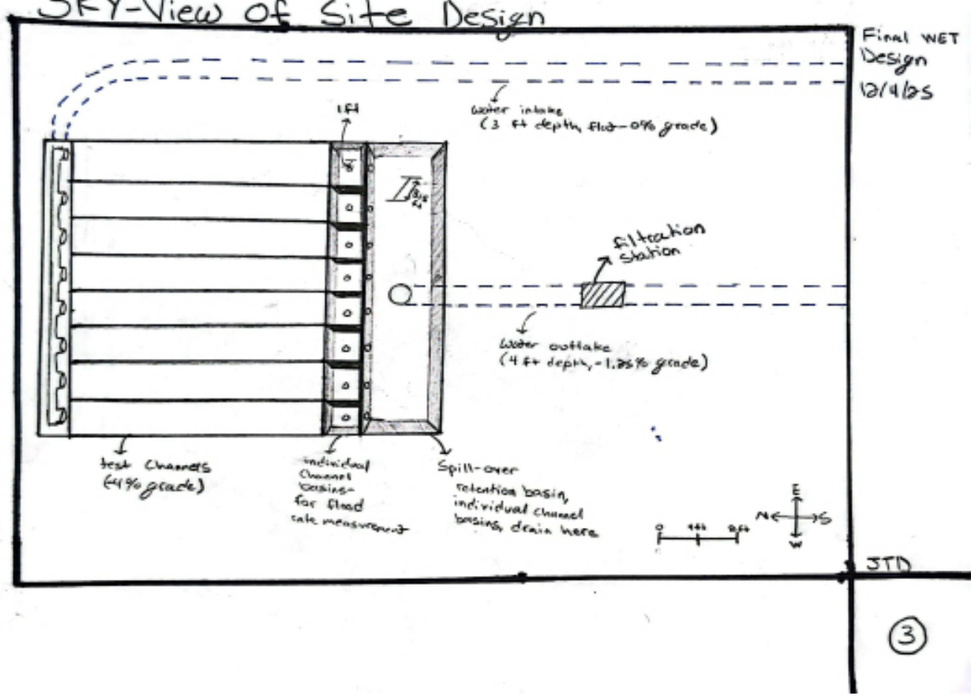


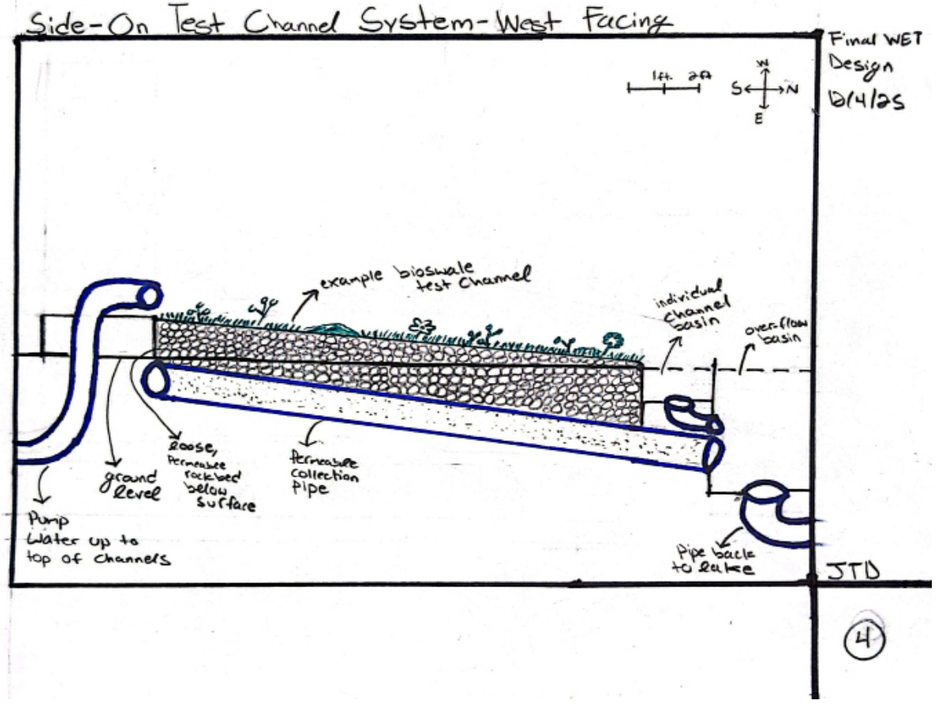
SKY-VIEW OF SITE DESIGN

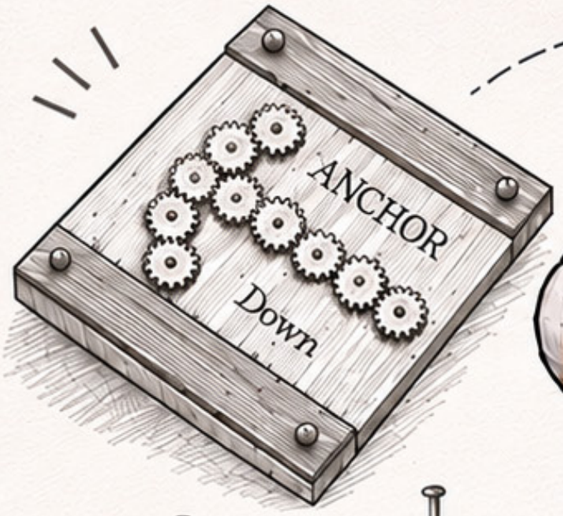


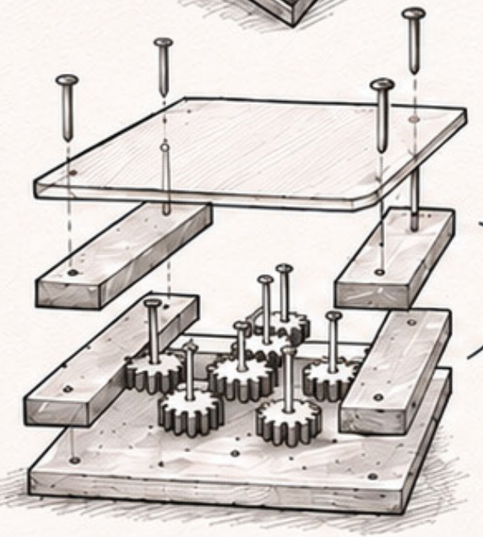

Lone Oaks Farm Area Overview
0 100ft 300ft
N
W E
S
Cub Creek
total area of construction, including pipes
Final WET Design 12/4/25
JTD
①

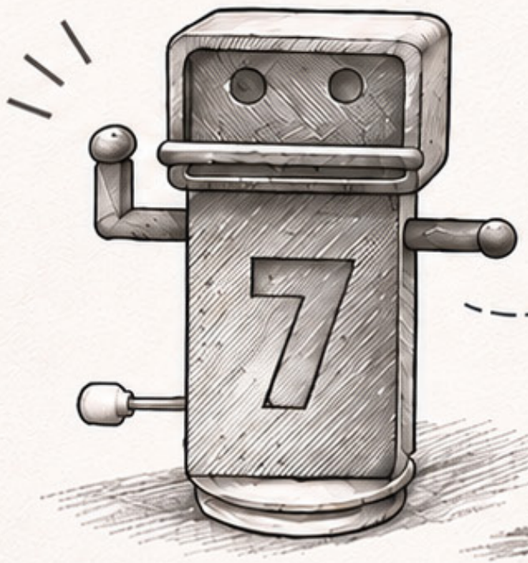

WET Center Site Plan Overview
N
W E
S
30ft 60ft
Proposed Site
water pipe & flow direction
Final WET Design 12/4/25
JTD
②


Sky-View of Site Design
1ft
water intake (3 ft depth, flat-0% grade)
filtration station
water outtake (4 ft depth, -1.25% grade)
test channels (4-9% grade)
individual channel basins- for flood rate measurement
Spill-over retention basin, individual channel basins drain here
0 4ft 8ft
N E S W
Final WET Design 12/4/25
JTD
③


Side-On Test Channel System-West Facing
1ft. 2ft
W
S N
E
example bioswale test channel
individual channel basin
overflow basin
ground level
loose, permeable rock bed below surface
permeable collection pipe
Pump water up to top of channels
Pipe back to lake
Final WET Design 12/4/25
JTD
④

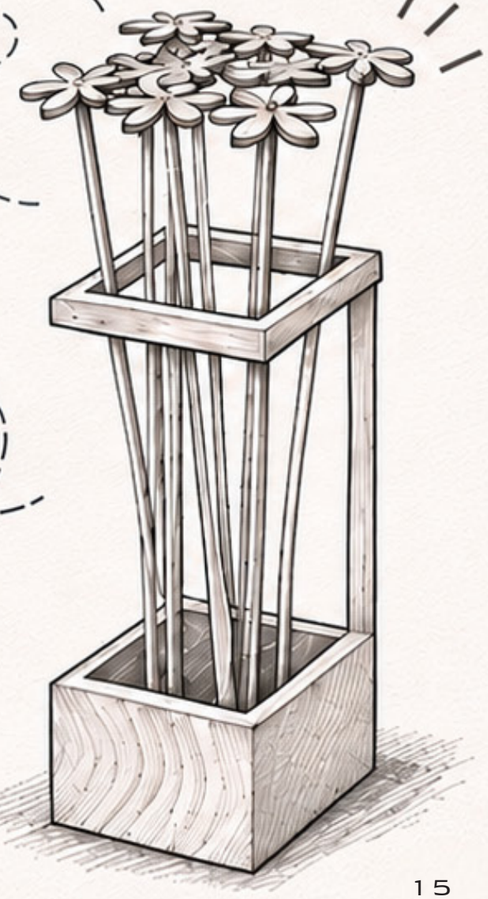




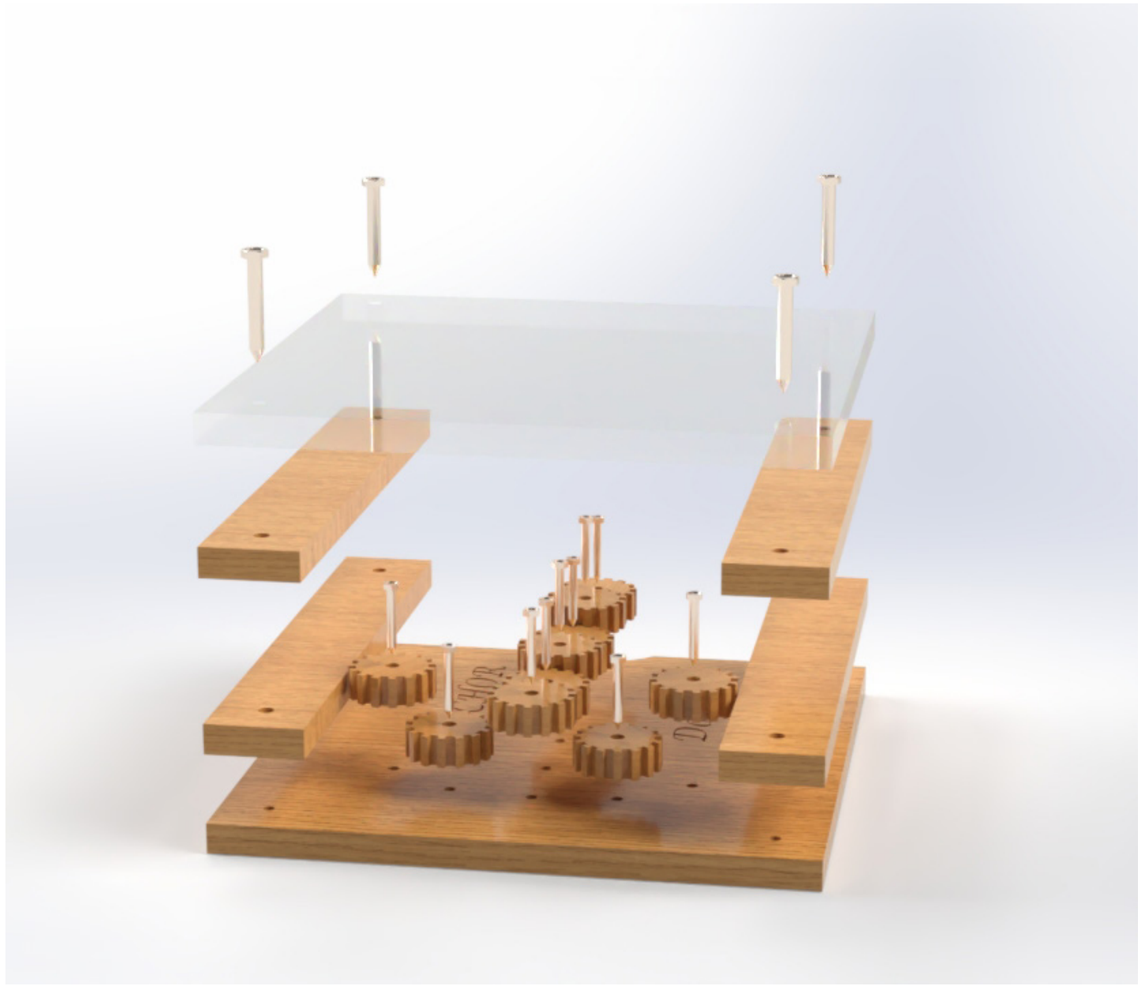


ISLAND OF MISFIT TOYS

SMALL PERSONAL PROJECTS DRIVEN BY
MY CURIOSITY AND LOVE FOR DESIGN.



Anchor Down



Everyone needs a little school pride. Inspired by Vanderbilt's history, I designed a coaster featuring interlocking, spinning gear heads that bring the university's symbol to life while adding a fidgit to my coffee table.

ANCHOR
Down

ANCHOR
Down

Blocky Football

My Favorite video game as a kid was Blocky Football. I wanted to bring the game to my desk and recreate it in real life. This led to a small passion project to design and create custom 3D-printed spin-top toys that remind me of a simpler time.

Isle of Skye

Everyone's name tells a story. My girlfriend's name traces back to her family's Scottish heritage on the Isle of Skye. Inspired by this connection, I designed a 3D-printable vase featuring a hand-sketched engraving of the island that wraps around its surface. The design blends personal meaning with digital fabrication, bringing a piece of her family's heritage into her everyday living space.

NEGATIVE DESIGN

DURING MY TIME AT *NEGATIVE DESIGN CO*, I EXPLORED SMALL-SCALE PRODUCT DESIGN ROOTED IN THE COMPANY'S PHILOSOPHY OF UTILIZING NEGATIVE SPACE. ONE OF MY FAVORITE CONCEPTS WAS A VASE THAT SHIFTS THE FOCUS FROM THE VESSEL ITSELF TO THE SPACE IT CREATES.